Rhizobium

Bio-fertilisers containing rhizobium bacteria are called *rhizobium* bio-fertilizer culture. Symbiotic bacteria that reside inside the root nodules convert the atmospheric nitrogen into a bio available form to the plants. This nitrogen fixing bacterium when applied to the soil undergoes multiplication and fixes the atmospheric nitrogen in the soil. *Rhizobium* is best suited for the paddy fields which increase the yield by 15 – 40%.

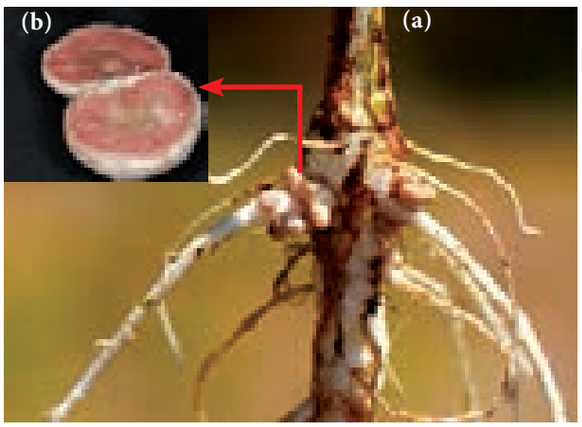


Figure 9.4 (a) : Root nodules occur on root
(b) C.S. of Root nodule

Azolla

Azolla is a free-floating water fern that fixes the atmospheric nitrogen in association with nitrogen fixing blue green alga *Anabaena azolla*. It is used as a bio-fertilizer for wetland rice cultivation and is known to contribute 40 – 60 kg/ha/crop. The agronomic potential of *Azolla* is quite significant particularly for increasing the yield of rice crop, as it quickly decompose in soil.



Figure 9.5: (a) *Azolla* in paddy field
(b) *Azolla*

Arbuscular mycorrhizae

Arbuscular mycorrhizae (AM) is formed by the symbiotic association between certain phycomycetous fungi and angiosperm roots. They have the ability to dissolve the phosphates found in abundance in the soil.

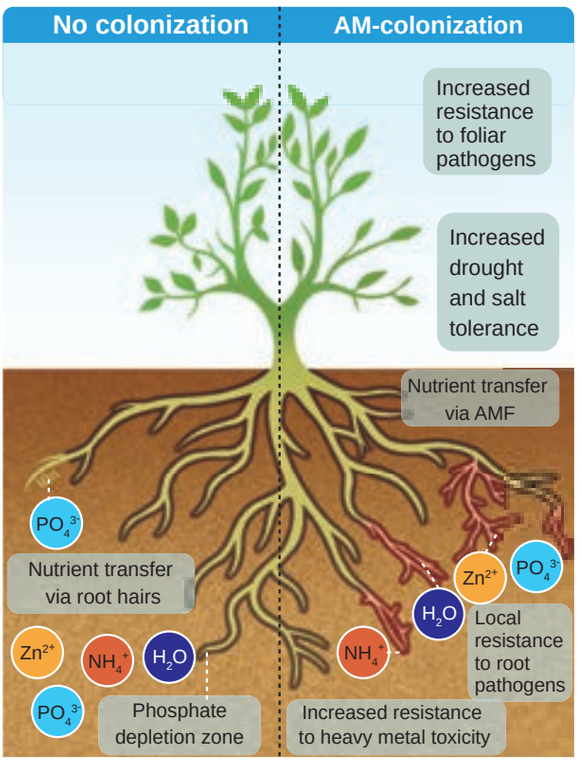


Figure 9.6 Benefits of AM colonisation

Apart from increasing the availability of phosphorus, AM provides necessary strength to resist disease, germs and unfavourable weather conditions. It also assures water availability.

Seaweed Liquid Fertilizer

Seaweed liquid fertilizer (SLF) contains cytokinin, gibberellins and auxin apart from macro and micro nutrients. Most seaweed based fertilizers are made from kelp(brown algae) which grows to length of 150 metres. seaweed Liquid fertilizer is not only organic but also eco-friendly. The alginates in the seaweed that react with metals in the soil and form long, cross-linked polymers in the soil. These polymers improve the crumbing in the soil, swell up when they get wet and retain moisture for a long time. They are especially



Figure 9.7 : Seaweed – Kelp

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